

AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 05

Claude AI for Productivity

Long Docs · Coding · Writing · Projects · API

Lesson 05 of 30 | AI & Productivity Program

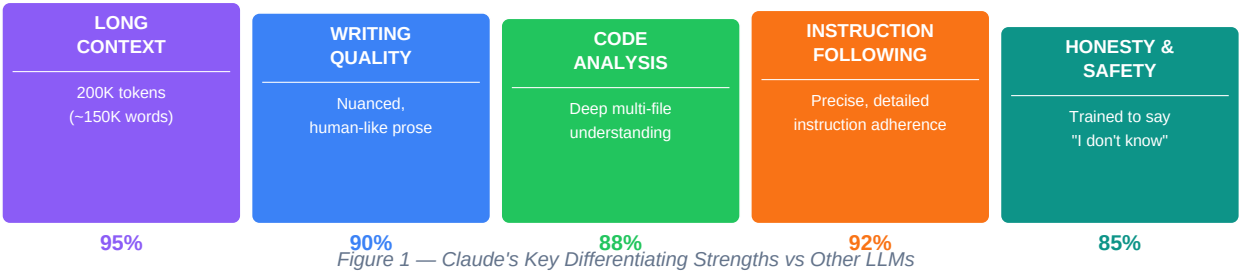
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- Claude's unique strengths vs ChatGPT, Gemini and other LLMs
- Claude Projects — persistent memory and custom instructions
- Analysing long documents (200K token context window)
- Coding with Claude — debugging, refactoring, explaining code
- 10 high-value productivity workflows you can use today

01 | What Makes Claude Different?

Claude is an AI assistant developed by **Anthropic** — an AI safety company founded in 2021 by former OpenAI researchers. Where ChatGPT aims for broad capability, Claude is optimised for **safety, honesty, and handling complex, nuanced tasks**. Claude is the AI of choice for professionals who need reliable, thoughtful assistance on high-stakes work.



Capability	Claude	ChatGPT-4o	Gemini 1.5 Pro	Best Use
Context window	200K tokens (~150K words)	128K tokens	1M tokens (but slower)	Claude for most professional documents; Gemini for huge data
Code understanding	Excellent — multi-file, deep analysis	Very good	Good	Claude for code review and refactoring; all good for generation
Long-form writing	Best nuance and tone control	Very good	Good	Claude for professional reports, articles, technical writing
Following complex instructions	Industry-leading precision	Very good	Good	Claude when instruction compliance is critical
Honesty about limits	Will say "I don't know" — trained for calibration	May confidently hallucinate	May confidently hallucinate	Claude when factual reliability matters most
Image understanding	Claude 3.5+ reads images, PDFs, diagrams	Yes (GPT-4o)	Yes	All three are capable; Claude handles technical diagrams well

Constitutional AI — Why Claude is Different

Claude is trained using Constitutional AI (CAI) — a method where the model learns from a set of principles rather than purely from human feedback. This makes Claude more consistent, less likely to give sycophantic answers, and more willing to push back on incorrect assumptions — which makes it a better thinking partner.

02 | Claude Projects — Persistent AI Context

Claude Projects (available on Claude.ai Pro/Team) lets you create a dedicated AI workspace with **persistent context, custom instructions, and uploaded documents** that Claude remembers across every conversation in that project. This is the closest thing to having a dedicated AI expert for each of your work domains.

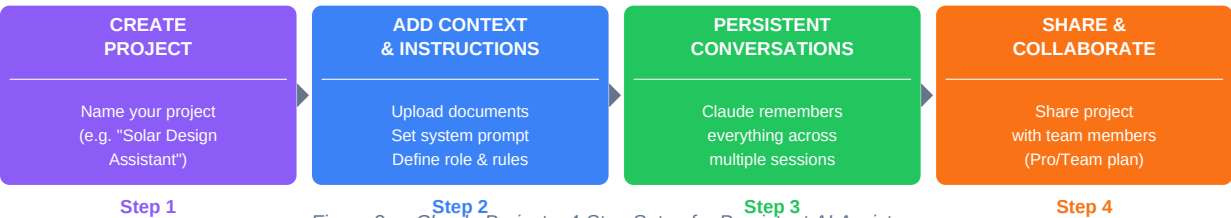


Figure 2 — Claude Projects: 4-Step Setup for Persistent AI Assistance

Project Type	Instructions to Include	Documents to Upload
Solar Design Assistant	"You are a solar PV engineer. Always use IEC standards. Show all calculations. Use metric units."	IEC 62109, project site data, client load profile, equipment specs
Technical Report Writer	"You are a technical writer. Always use formal engineering tone. Structure reports with Executive Summary, Body, Conclusion."	Company report template, style guide, previous reports as examples
Code Review Partner	"You are a senior software engineer. Point out bugs, security issues, and performance problems. Suggest best practices."	Codebase files, API documentation, coding standards guide
Study Assistant	"You are an exam tutor for [subject]. Ask me questions to test my understanding. Do not give answers immediately — make me think first."	Textbooks, past exam papers, lecture notes, syllabus
Business Analyst	"You are a business analyst. When given data, identify trends, outliers, and strategic implications. Format output as executive brief."	Financial reports, market research, competitor analysis data

03 | Analysing Long Documents with Claude

Claude's 200K token context window means you can paste or upload documents up to approximately 150,000 words — an entire technical manual, a 200-page contract, or an entire software codebase — and ask detailed questions about it. This is transformative for professionals who spend hours reading and summarising documents.

Document Type	What to Ask Claude	Time Saved
Technical standard (100+ pages)	"Summarise the sections relevant to solar PV installation safety. List all requirements for DC wiring." Then ask follow-up questions.	3–4 hours → 15 minutes
Legal contract / tender document	"Identify all payment terms, penalty clauses, and client obligations. Highlight any terms that are unusual or high-risk."	4–6 hours → 30 minutes

Research paper / academic report	"What is the methodology? What are the main findings? What are the limitations? How does this relate to [your topic]?"	1–2 hours → 10 minutes
Software codebase	"Explain how the authentication module works. Find any potential security vulnerabilities in this code."	2–4 hours → 20 minutes
Meeting transcript / notes	"Extract all action items and who they are assigned to. Summarise key decisions made."	45 min → 3 minutes

The Grounding Technique

When asking Claude about a specific document, always paste the document first, then ask your question. Add: "Answer only from the document I have provided. If you cannot find the answer in the document, say so." This dramatically reduces hallucination and keeps Claude on-topic.

04 | Coding and Technical Work with Claude

Claude is outstanding for technical and coding tasks — not just writing code, but **understanding, explaining, and improving existing code**. Unlike tools that only generate code snippets, Claude can hold entire multi-file systems in context and reason about their architecture.

Coding Task	Best Prompt Pattern	Example
Debug code	"This code should [expected]. Instead it [actual]. Here is the code: [paste]. What is wrong and how do I fix it?"	"This Python function should return the peak sun hours for a location. Instead it returns None. [paste code]"
Explain code	"Explain this code to me as if I am [experience level]. Focus on [specific aspect]. Use real examples."	"Explain this Revit API script to someone who knows Python but not Revit. Explain what each method does."
Refactor code	"Refactor this code to be [cleaner/faster/more readable]. Keep the same functionality. Explain each change you make."	"Refactor this 200-line function into smaller, well-named functions. Explain why you split it the way you did."
Write tests	"Write unit tests for this function. Cover: normal cases, edge cases, and error cases. Use [testing framework]."	"Write pytest tests for this solar yield calculator. Test with typical and edge case input values."
Architecture review	"Review this [API/database/system design]. Identify: potential bottlenecks, security issues, scalability problems."	"Review this Flask API structure and identify any security vulnerabilities or poor practices."

05 | 10 High-Value Claude Workflows

#	Workflow	Prompt Pattern
1	Daily email processing	"I have [N] emails. Categorise each as [action/FYI/delete]. Draft reply for the action items. [paste emails]"
2	Meeting preparation	"I have a meeting with [person/organisation] about [topic]. Prepare 5 questions I should ask and 3 potential objections to prepare for."
3	Report first draft	"Write a technical report section on [topic] for [audience]. Use formal tone. Include: intro, key points, recommendations. 400 words."
4	Decision analysis	"I must decide between [A] and [B]. Pros and cons of each. Which do you recommend and why? Be honest about uncertainty."
5	Learning accelerator	"I have 30 minutes to understand [topic]. What are the 5 most important concepts? Give one real example for each."
6	Document comparison	"Compare Document A [paste] vs Document B [paste]. List key differences in a table."
7	Data interpretation	"Here is [table/data]. What are the 3 most important insights? What action should I take based on this?"
8	Proposal writing	"Write an executive summary for a proposal to [client] for [service]. Budget: [X]. Duration: [Y]. Key benefit: [Z]."
9	Risk identification	"I am planning to [project/decision]. What are the top 10 risks? Rate each 1-5 for likelihood and impact. Suggest mitigations."

10	Personal development	"I want to improve at [skill] in [timeframe]. Create a weekly learning plan with resources and practice exercises."
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What comes next — AI-006

Google Gemini AI — how Gemini 1.5 Pro compares to Claude, its 1-million token context window, integration with Google Workspace (Docs, Sheets, Gmail), Gemini Advanced, and NotebookLM for research.